Student guide | Lesson 1 of 5

A square that moves

Make a window and steer a square with the arrow keys.

Name: _____ Date: _____

Today you will make

A window with a green square that you can steer with the arrow keys. It is the start of your own Snake game.

Words you will meet

Word	What it means
module	A toolbox of ready-made code. *turtle* is the toolbox for drawing and simple games.
variable	A name that holds a value, like STEP = 20.
function	A named set of instructions. You make one with **def**.
object	Something we create from a toolbox, like our square.
event	Something that happens, like pressing a key.

Before you start

- Open IDLE (or your editor) and choose **File > New File**.
- Save it as **snake.py**. Make a new folder for this course so you can find it.
- Type the code yourself instead of copying. You learn more from your typos.

Step 1: Open a window

Type this at the top of your file. **screen.mainloop()** must stay the very last line. Everything new goes above it.

```
import turtle

STEP = 20

screen = turtle.Screen()
screen.title("Snake")
screen.bgcolor("black")
screen.setup(600, 600)

screen.mainloop()
```

What it does

- **import turtle** loads the turtle toolbox.
- **STEP = 20** is how many pixels the square moves each time.
- The **screen** lines make a 600 by 600 black window called Snake.
- **mainloop()** keeps the window open and waiting for keys.

Try it: Press F5 to run. You should see an empty black window. Close it to stop.

Step 2: Make the head

Add this **above screen.mainloop()**:

```
head = turtle.Turtle("square")
head.color("lime")
head.penup()
```

What it does

- **turtle.Turtle("square")** creates the object, our snake's head.
- **color("lime")** makes it bright green.
- **penup()** lifts the pen so the square does not draw a line when it moves.

> **Try it:** Run it. A green square sits in the middle.

Step 3: Write four movement functions

Add these **above screen.mainloop()**. Each indented line belongs to the function above it, so use four spaces.

```
def go_up():
    head.sety(head.ycor() + STEP)

def go_down():
    head.sety(head.ycor() - STEP)

def go_left():
    head.setx(head.xcor() - STEP)

def go_right():
    head.setx(head.xcor() + STEP)
```

What it does

- **ycor()** is the square's up/down position. **sety()** sets a new one.
- Adding STEP moves it up. Taking STEP away moves it down.
- **xcor()** and **setx()** do the same left and right.

> **Try it:** Run it. Nothing moves yet. The functions exist but nothing calls them. That comes next.

Step 4: Connect the keys

Add these **above screen.mainloop()**, after your functions:

```
screen.listen()
screen.onkeypress(go_up, "Up")
screen.onkeypress(go_down, "Down")
screen.onkeypress(go_left, "Left")
screen.onkeypress(go_right, "Right")
```

What it does

- **screen.listen()** tells the window to pay attention to the keyboard.
- **onkeypress(go_up, "Up")** means: when Up is pressed, run go_up.
- Notice there are no brackets after go_up. We pass the function itself, not the result of running it.

> **Try it:** Run it, click on the turtle window, and use the arrow keys. The square should move.

Think about it: What would happen if you deleted head.penup()? Test it, then put it back.

Your answer: _____

Think about it: What happens if you change STEP to 50?

Your answer: _____

Challenges (pick one, or do them all)

- [ ] Change the square's colour. Try "red", "orange" or "hotpink".
- [ ] Change the shape: try "circle", "triangle" or "arrow".
- [ ] Make the square move faster or slower by changing STEP.
- [ ] Change the window's background colour.

Stuck? Check here first

What you see	Likely cause and fix
Keys do nothing	Click on the turtle window first. Check that screen.listen() is there.
IndentationError	Lines inside a def must start with four spaces.
NameError: name 'turtl' is not defined	A typo. Compare the word in the error with the code.
The window closes straight away	screen.mainloop() is missing from the end.

I can...

- [ ] import a module.
- [ ] create an object and change how it looks.
- [ ] write a function with def.
- [ ] connect a key press to a function.

Check your work: the complete code

Compare your file with this one, line by line. If yours does not run, the mistake is usually a typo, a missing colon, or indentation that is not lined up.

```
import turtle

STEP = 20

screen = turtle.Screen()
screen.title("Snake")
screen.bgcolor("black")
screen.setup(600, 600)

head = turtle.Turtle("square")
head.color("lime")
head.penup()

def go_up():
    head.sety(head.ycor() + STEP)

def go_down():
    head.sety(head.ycor() - STEP)

def go_left():
    head.setx(head.xcor() - STEP)

def go_right():
    head.setx(head.xcor() + STEP)

screen.listen()
screen.onkeypress(go_up, "Up")
screen.onkeypress(go_down, "Down")
screen.onkeypress(go_left, "Left")
screen.onkeypress(go_right, "Right")

screen.mainloop()
```